

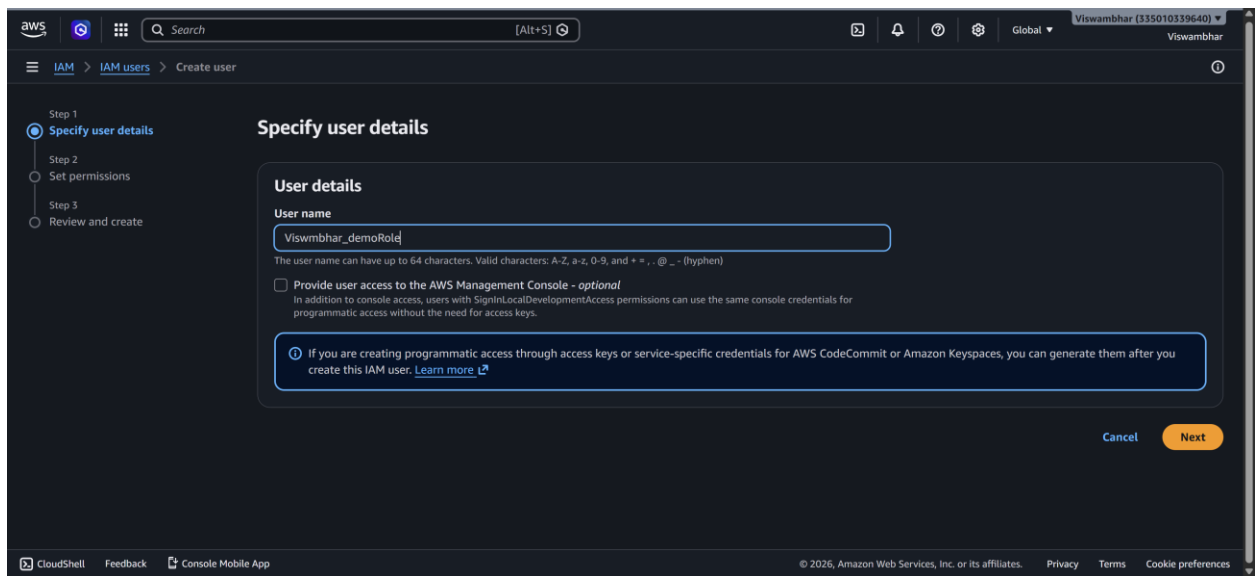
WEEK-12 –Creating IAM USER with permission Access it with GUI and CLI

Part A: GUI Access (Management Console)

Objective: Create an IAM user with restricted permissions and verify access via the AWS web portal.

1. Administrative Setup (Root User)

- **Step 1:** Log in to the AWS Management Console as the **Root User**.
- **Step 2:** Search for **IAM** under the "Security, Identity, & Compliance" category.
- **Step 3:** Navigate to Users and click Create user.



The screenshot shows the AWS IAM 'Specify user details' page. The breadcrumb navigation is 'IAM > IAM users > Create user'. On the left, a progress bar shows three steps: 'Specify user details' (selected), 'Set permissions', and 'Review and create'. The main section is titled 'Specify user details' and contains a 'User details' box. Inside this box, there is a 'User name' field with the value 'Viswambhar_demoRole'. Below the field, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, -, @, _ (hyphen)'. There is an unchecked checkbox labeled 'Provide user access to the AWS Management Console - optional'. Below this checkbox, a note says: 'In addition to console access, users with SigninLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.' At the bottom of the box, there is a tip: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)'. At the bottom right of the page, there are 'Cancel' and 'Next' buttons. The footer includes 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information for Amazon Web Services.

- **Step 4:** User Details: Enter a name (e.g., S3_Specialist). Check the box for "Provide user access to the AWS Management Console." – custom password.

User name
Viswmbhar_demoRole
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, -, @, _ (hyphen)

☒ **Provide user access to the AWS Management Console - optional**
In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ **Custom password**
Enter a custom password for the user.

☐ Show password

☒ **Users must create a new password at next sign-in - Recommended**
Users automatically get the `IAMUserChangePassword` policy to allow them to change their own password.

ⓘ If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel Next

- **Step 5: Set Permissions:** Select "Attach policies directly." Search for and select **AmazonS3FullAccess**.

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1484)
Choose one or more policies to attach to your new user.

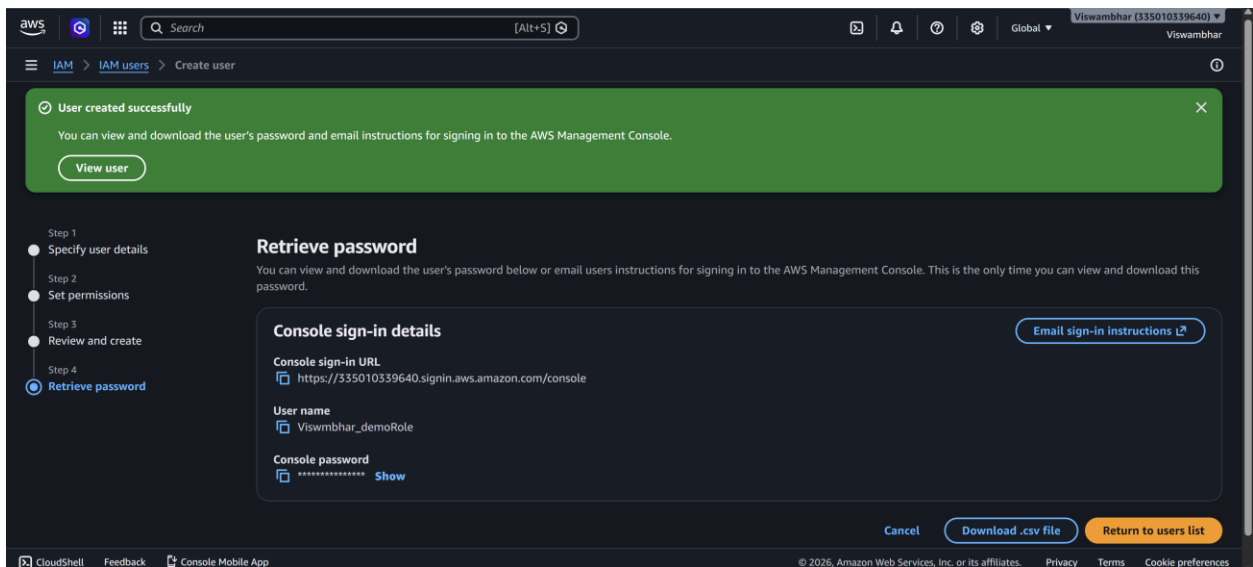
Search: AmazonS3FullAccess Filter by Type: All types 1 match

☒	Policy name	Type	Attached entities
☒	AmazonS3FullAccess	AWS managed	0

▶ Set permissions boundary - optional

Cancel Previous Next

- **Step 6: Review & Create:** Complete the wizard. On the final page, click Download .csv file.

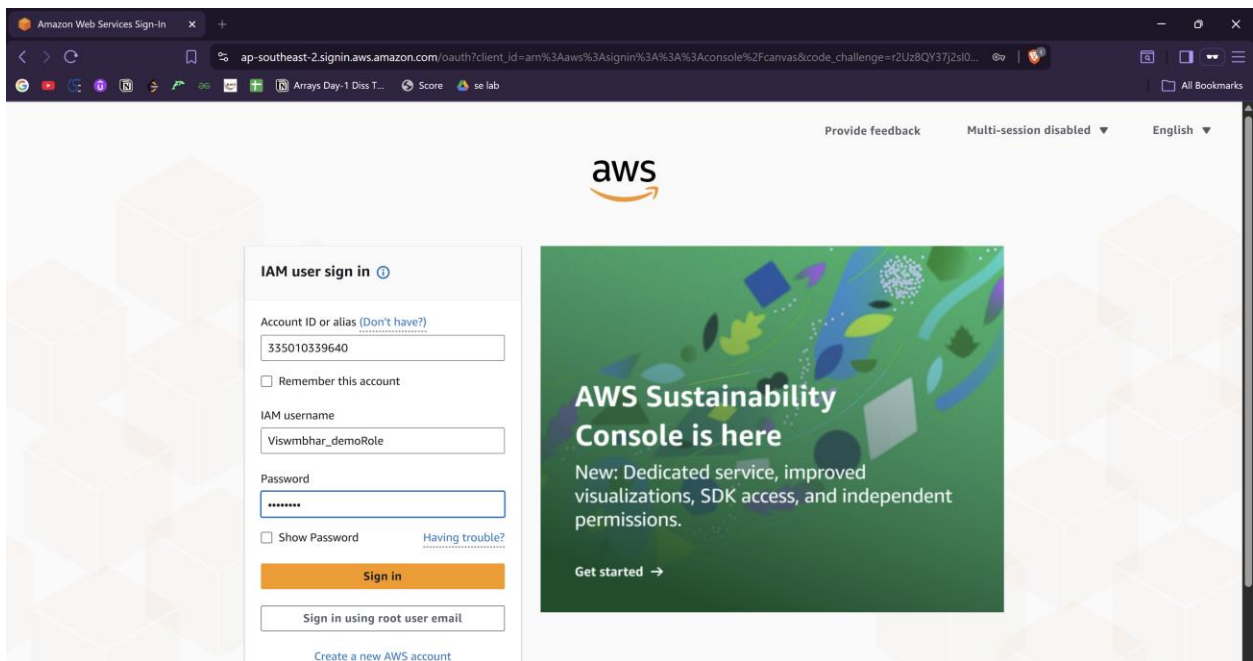


Note: This file contains your Console Password and the unique Sign-in URL.

- **Step 7:** Copy your 12-digit Account ID (found in the top-right dropdown or the .csv) and Sign Out.

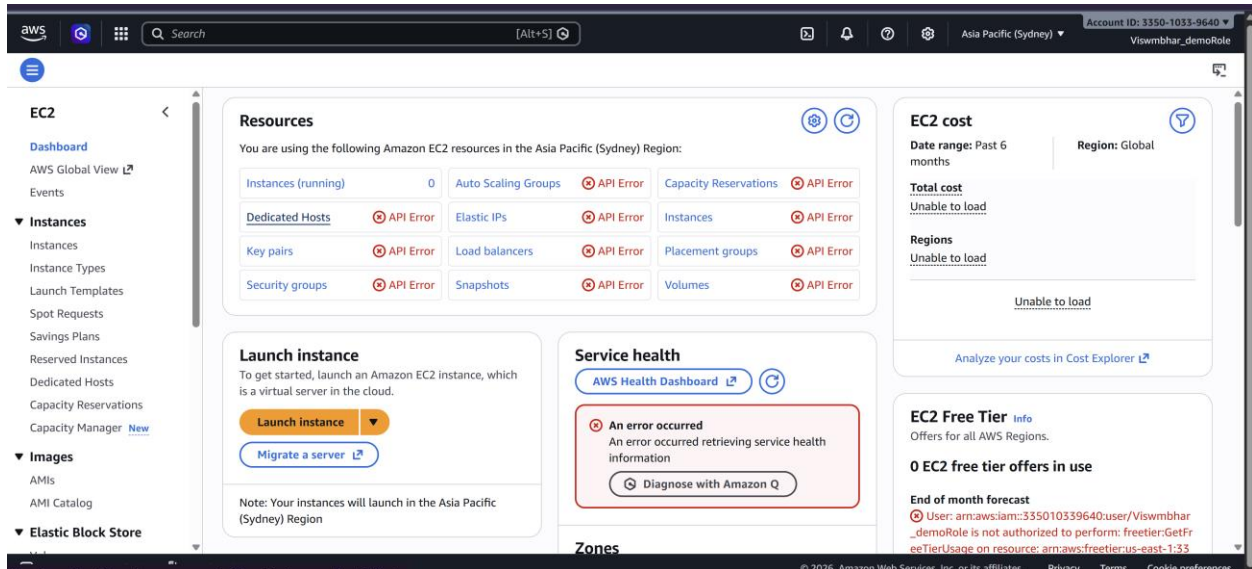
2. Verification (IAM User Login)

- **Step 8:** Use the Sign-in URL from the .csv (or go to the AWS sign-in page and select **IAM User**).
- **Step 9:** Enter the **Account ID**, then the **IAM Username** and **Password** from the .csv.

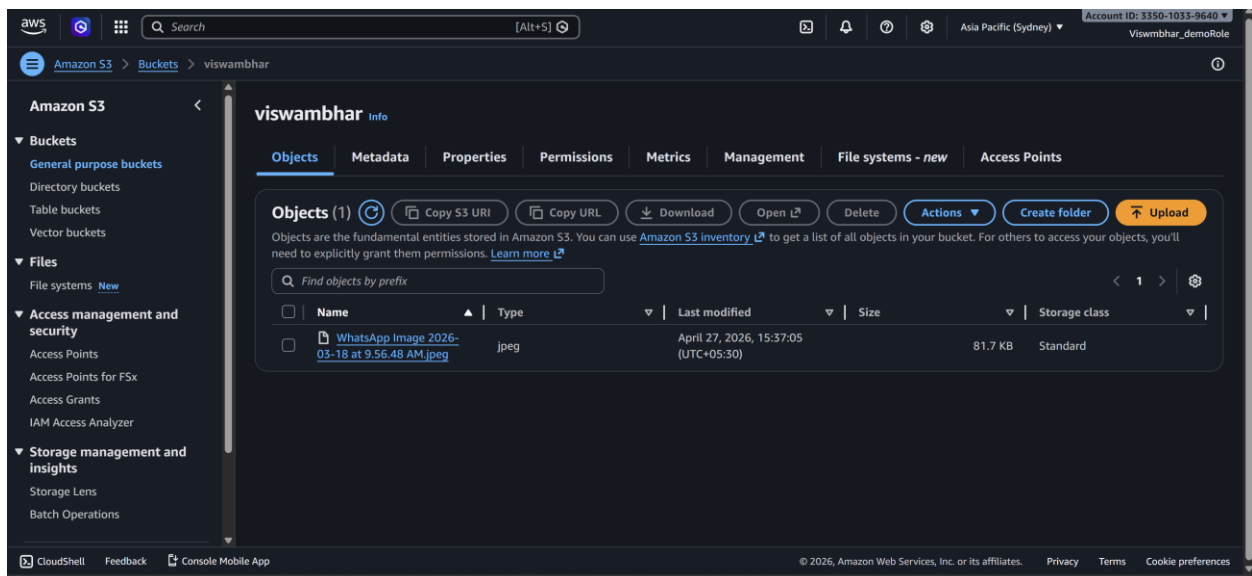


- **Step 10: The Logic Test:**

- **Test 1 (Failure):** Navigate to **EC2**. Try to view instances. You will see "API Error" or "Access Denied." This proves the user is restricted.



- **Test 2 (Success):** Navigate to **S3**. Create a new bucket. It will work perfectly because of the attached policy.

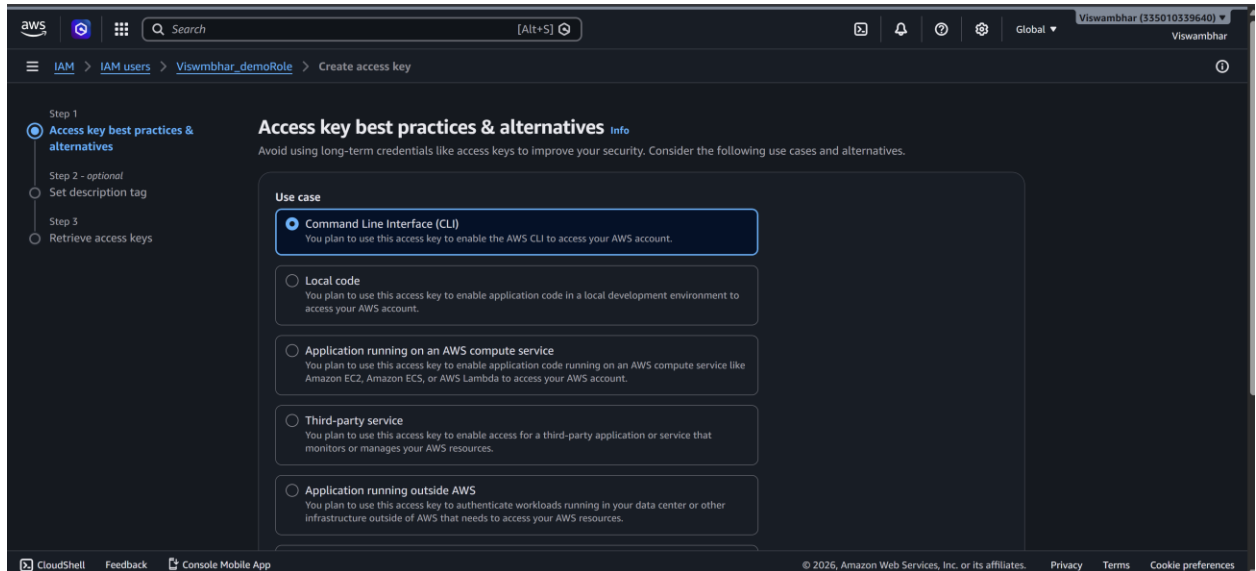


Part B: CLI Access (Programmatic Access)

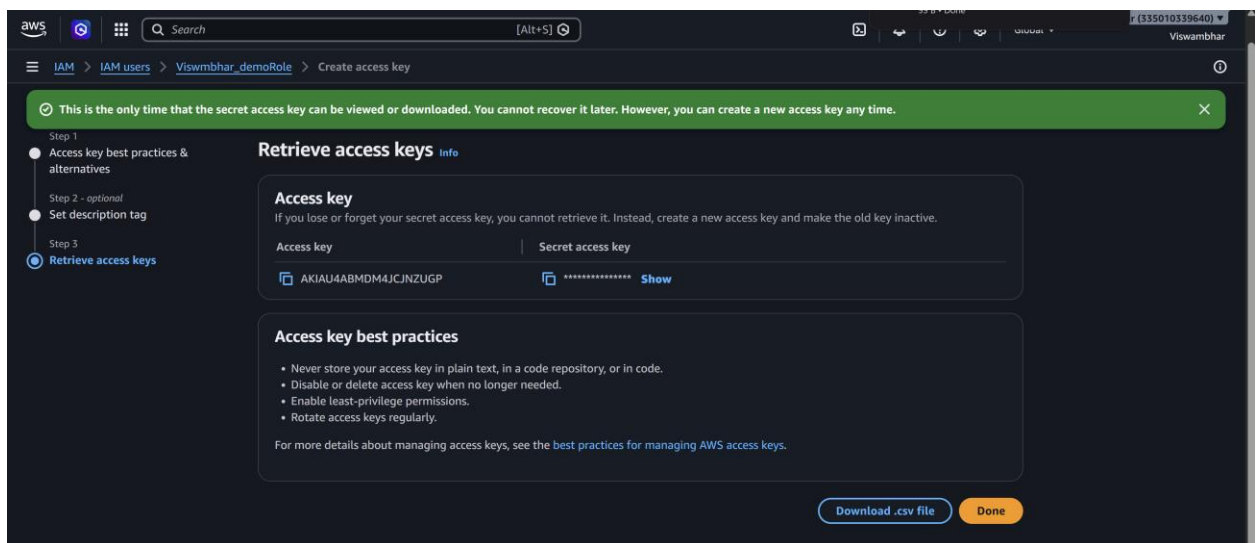
Objective: Access AWS resources from a standalone system (CMD/Terminal) using security keys.

1. Credential Generation

- **Step 1:** While logged in as an Admin/Root, go to **IAM > Users** and click on your user.
- **Step 2:** Go to the **Security credentials** tab. Scroll to **Access keys** and click **Create access key**.
- **Step 3:** Select **Command Line Interface (CLI)**. Acknowledge the warning and click **Next**.

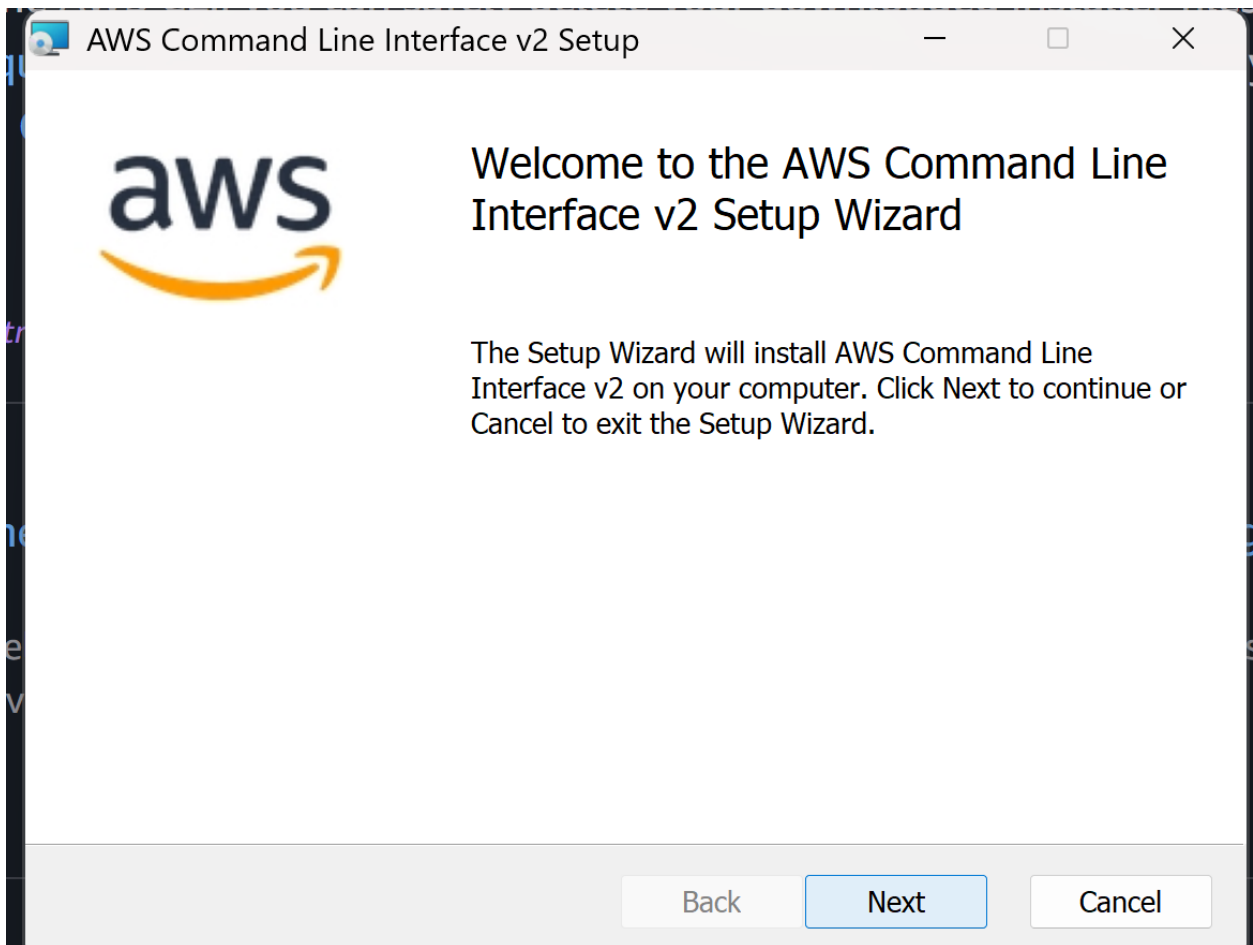


- **Step 4:** Download the .csv file. > **Critical:** This contains the **Access Key ID** and **Secret Access Key**. If lost, you must delete and recreate them.



2. Standalone System Configuration

- **Step 5:** Install the **AWS CLI Tool** from google on your computer.



- **Step 6:** Open your Command Prompt (CMD) or Terminal.
- **Step 7:** Type `aws configure` and press Enter.
- **Step 8:** Provide the following from your .csv:
 - **Access Key ID**
 - **Secret Access Key**
 - **Default region:** (e.g., ap-south-1)
 - **Default output:** json

```
C:\Users\LENOVO>aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws log
in'.

AWS Access Key ID [None]: AKIAU4ABMDM4JCJNZUGP
AWS Secret Access Key [None]: 6OSS1NdnRD9y4BQMgwkC4lCcH/QH260draFQirTg
Default region name [None]:
Default output format [None]:

C:\Users\LENOVO>aws configure
AWS Access Key ID [*****ZUGP]:
AWS Secret Access Key [*****irTg]:
Default region name [None]: us-east-1
Default output format [None]: json

C:\Users\LENOVO>s3 ls
```

3. Verification via Commands

- **Step 9:** Run `aws s3 ls`. This should list your buckets.

```
C:\Users\LENOVO>aws s3 ls
2026-04-27 15:45:56 viswambhar

C:\Users\LENOVO>
```

- **Step 10:** Run `aws s3 mb s3://your-unique-bucket-name`. (Success)

```
C:\Users\LENOVO>aws s3 mb s3://viswambhar-test-2
make_bucket: viswambhar-test-2

C:\Users\LENOVO>
```

- **Step 11:** Run `aws iam list-users`.
 - **Result:** You will get an **(AccessDenied)** error.

```
aws: [ERROR]: An error occurred (AccessDenied) when calling the ListUsers operation: User: arn:aws:iam::335010339640:use
r/Viswambhar_demoRole is not authorized to perform: iam:ListUsers on resource: arn:aws:iam::335010339640:user/ because no
identity-based policy allows the iam:ListUsers action

C:\Users\LENOVO>
```

This confirms that the CLI is respecting the same IAM policy as the GUI.